

Tools used

Claude (chat assistant). I used it for the project plan, PowerShell and git commands, the event engine and statistics code, and help reading the outputs. I ran all the code myself and checked the outputs at each step.

How I used it

I split the assignment into small steps: data, validation, engine, statistics, robustness grid, out-of-sample test, falsification and backtest. AI wrote the first version of `src/engine.py` and `src/research_stats.py`, and I ran and tested them in notebooks. I also asked it to explain confidence intervals, p-values and the robustness grid.

My own decisions

- 1) I chose to keep the 2008 and 2020 crash data, because they are real events and removing them would delete exactly the falls I am studying
- 2) I fixed the primary test (-2%, 5-day hold) before seeing results, and refused to pick the best-looking cell of the grid. With 12 tests, one lucky result is expected by chance
- 3) I froze the out-of-sample period (2018 onward), wrote the pass/fail rule before running it, and ran it once without changing any setting afterwards
- 4) I treat the falsification checks as diagnostics only, because I ran them after seeing the out-of-sample result. The "excluding 2008 and 2020" result is post-hoc, and I do not claim it as a finding

Suggestions I changed or disagreed with

- The first exit definition was ambiguous, so I rewrote it: the entry day counts as day 1
- The AI suggested a proper backtest, but the research showed no edge, so I labelled the backtest a demonstration only

Incorrect AI output I found

1. "mkdir data notebooks src docs" failed in PowerShell, which needs commas between folder names
2. "pip freeze > requirements.txt" saves the file as UTF-16 in PowerShell, so I re-saved it as UTF-8
3. The git remote command was given with a placeholder username (YOUR_USERNAME), which I had to replace with my own
4. The AI flagged the last data row (2026-09-23, volume 0) as a possibly incomplete bar, and I dropped it. The missing 2026-09-22 is still unverified

What I learned

A negative result is a valid result. Statistical significance is different from trading significance, and even the best-looking grid cells had confidence intervals too wide to trust. Overlapping events and clustered crashes make a small sample look more informative than it is. AI-generated code and commands need to be checked, because several suggestions failed or were ambiguous.